DANIEL KARIMI

DATA SCIENTIST (ANALYTICS & ML)

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ABOUT ME

Highly analytical engineering student at **Universidad Politécnica de Madrid** (expected graduation 2025/2026), specializing in turning complex, large-scale datasets into actionable insights. I possess a robust mathematical and statistical foundation, complemented by practical experience architecting scalable data extraction and analysis pipelines to solve real-world analytical bottlenecks.

During my tenure as an AI Engineer Intern at CSIC, I conducted rigorous deep-dive analyses and root cause investigations on massive datasets. I am highly proficient in modern Data Science scripting languages and environments, particularly **Python (Pandas, NumPy, Scikit-learn) and MATLAB**, which I have extensively used for statistical modeling, exploratory data analysis, and uncovering hidden data patterns.

I am eager to bring my passion for building reproducible analytical models and utilizing **Machine Learning and statistical techniques** to the Amazon Books team. My goal is to apply data-driven problem-solving to ensure unmatched catalog quality and improve the book discovery experience for millions of customers.

TECHNICAL EXPERTISE

📈 DATA SCIENCE & ANALYTICS

Statistical Modeling • Exploratory Data Analysis • Root Cause Analysis • Data Mining • Data Quality Monitoring • Quantitative Reporting

🛠️ SCRIPTING & TOOLS

Python • MATLAB • R (Familiarity) • SQL Concepts • Pandas / NumPy • Jupyter Notebooks

🤖 MACHINE LEARNING

Predictive Modeling • Deep Learning Pipelines • PyTorch / TensorFlow • Algorithm Optimization • Model Evaluation Metrics • Agentic Technology

💻 SOFTWARE / SYSTEMS

AWS (S3, EC2 Familiarity) • Linux / Unix OS • Git & Version Control • Bash & Automation • Data Pipeline Architecture • Scalable Code Design

📊 COMMUNICATION

Data Visualization • Technical Documentation • Stakeholder Presentation • Scientific Reporting • Actionable Insights

🔬 MATHEMATICAL FOUNDATIONS

Linear Algebra • Advanced Calculus • Probability & Statistics • Signal Processing • Error/Deviation Analysis

Languages: Spanish (Native) • English (B2 – Aptis ESOL, British Council - capable of communicating complex concepts effectively)

PROFESSIONAL EXPERIENCE

Data Scientist & AI Intern | CSIC – Institute of Physical and Information Technologies (ITEFI)

Sep 2025 – Feb 2026

- Designed and implemented **scalable data extraction and analysis pipelines** using Python to process massive datasets, ensuring high-quality, normalized data tracking over time.
- Conducted **deep-dive exploratory data analyses** and deviation analysis to identify patterns and anomalies within raw scientific data, enabling data-driven architectural decisions.
- Built and evaluated analytical prototypes using Machine Learning (PyTorch) and statistical modeling to automate data curation, drastically reducing manual analysis time from hours to minutes.
- Performed rigorous root cause analysis of model inaccuracies by diving deep into parameters affecting ML performance, optimizing predictive outcomes.
- Collaborated tightly with cross-functional scientific teams, translating complex analytical findings and statistical outputs into clear, reproducible documentation and actionable insights.

Python | Deep-Dive Analysis | Data Mining | Statistical Modeling | Machine Learning | Root Cause Analysis

EDUCATION

Telecommunications Systems Engineering | **Universidad Politécnica de Madrid (ETSIT-UPM)** *Sep 2020 – Jul 2025*

- **Mathematical & Statistical Foundation:** Advanced coursework in Probability, Statistics, Algebra, and error deviation analysis, providing a rigid framework for complex analytical problem-solving.
- **Data Scripting & Algorithms:** Extensive use of data scripting languages including **MATLAB** (signal processing, statistical analysis, matrix operations) and **Python** for algorithm design.
- **Engineering & Architecture:** Scalable systems design, data structuring, and network architecture, fostering an environment of technical rigor and operational excellence.

MATLAB | Python | Statistics | Algorithms | Mathematical Modeling

Technology Baccalaureate, **IES Isabel la Católica** *2018 – 2020*

Science and Technology track. Honours. Strong foundation in mathematics and physics.

CERTIFICATIONS

AI and Productivity (Google): Efficient use of AI tools to automate data extraction workflows. •

Generative AI (MIT): Implementation of GenAI solutions and agentic technologies for data synthesis. •

Internet of Things (MIT): IoT data pipelines, large-scale sensory data collection, and applied statistical models. •

Aptis ESOL B2 (British Council): Official English certification. Proficient in effectively communicating complex concepts through both written and verbal channels.

KEY PROJECTS

CellTracker (Analytical Prototype) MACHINE LEARNING PIPELINE

PYTHON • STATISTICAL MODELING • DATA VISUALIZATION

Algorithmic tracking framework built for deep data analysis.

Architected an end-to-end Machine Learning pipeline combining custom models with robust statistical algorithms (Kalman Filters) to process and analyze complex data parameters. Designed logic to quantify tracking accuracy and build scalable tracking components.

- Wrote comprehensive analytical scripts to monitor data deviations and validate hypotheses.
- Generated advanced data visualizations to translate complex performance outputs into clear, presentation-ready insights.

MigraCell Accelerator DATA EXTRACTION

PYTHON • AUTOMATION • DEEP ANALYSIS

Automated analysis tool that quantifies previously manual assessments.

Developed an analytical prototype utilizing Machine Learning for large-scale data segmentation. Programmed data extraction pipelines that drastically improved monitoring of analytical variables and tracked operational improvements.

- Applied deep learning models in an applied environment to solve real-world problems.
- Engineered reproducible scripts to ensure long-term data quality and analytical consistency.